

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 2: Polynomials

Step-by-step solutions to the textbook exercises: 8 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 2.1

Q1 Six graphs of $y = p(x)$ are given in the textbook (Fig. 2.10). Find the number of zeroes of $p(x)$ in each case.

SOLUTION

The zeroes of $p(x)$ are the x -coordinates of the points where the graph of $y = p(x)$ meets the x -axis. So count the points where each graph cuts or touches the x -axis.

Graph	What the graph does	Number of zeroes
(i)	A straight line parallel to the x -axis, above it. It never meets the x -axis.	0
(ii)	A curve that stays below the x -axis on the left and crosses it once on the right.	1
(iii)	A curve that crosses the x -axis at three points.	3
(iv)	A U-shaped curve that crosses the x -axis at two points.	2
(v)	A wavy curve that crosses the x -axis at four points.	4
(vi)	A curve that crosses the x -axis once and touches it at two more points.	3

Answer: (i) 0 (ii) 1 (iii) 3 (iv) 2 (v) 4 (vi) 3

2 Exercise 2.2

Q1 (i) Find the zeroes of $x^2 - 2x - 8$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$x^2 - 2x - 8 = x^2 - 4x + 2x - 8 = x(x - 4) + 2(x - 4) = (x - 4)(x + 2)$. The zeroes are 4 and -2 .

Sum of zeroes = $4 + (-2) = 2 = \frac{-(-2)}{1} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$.

Product of zeroes = $4 \times (-2) = -8 = \frac{-8}{1} = \frac{\text{constant term}}{\text{coefficient of } x^2}$.

Answer: zeroes 4 and -2 . Verified.

Q1 (ii) Find the zeroes of $4s^2 - 4s + 1$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$4s^2 - 4s + 1 = (2s - 1)^2$. Both zeroes are $\frac{1}{2}$.

Sum of zeroes = $\frac{1}{2} + \frac{1}{2} = 1 = \frac{-(-4)}{4}$.

Product of zeroes = $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = \frac{\text{constant term}}{\text{coefficient of } s^2}$.

Answer: zeroes $\frac{1}{2}$ and $\frac{1}{2}$. Verified.

Q1 (iii) Find the zeroes of $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

In standard form the polynomial is $6x^2 - 7x - 3 = 6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (3x + 1)(2x - 3)$. The zeroes are $-\frac{1}{3}$ and $\frac{3}{2}$.

Sum of zeroes = $-\frac{1}{3} + \frac{3}{2} = \frac{7}{6} = \frac{-(-7)}{6}$.

Product of zeroes = $-\frac{1}{3} \times \frac{3}{2} = -\frac{1}{2} = \frac{-3}{6}$.

Answer: zeroes $-\frac{1}{3}$ and $\frac{3}{2}$. Verified.

Q1 (iv) Find the zeroes of $4u^2 + 8u$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$4u^2 + 8u = 4u(u + 2)$. The zeroes are 0 and -2.

Sum of zeroes = $0 + (-2) = -2 = \frac{-8}{4}$.

Product of zeroes = $0 \times (-2) = 0 = \frac{0}{4}$, since the constant term is 0.

Answer: zeroes 0 and -2. Verified.

Q1 (v) Find the zeroes of $t^2 - 15$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$t^2 - 15 = (t - \sqrt{15})(t + \sqrt{15})$. The zeroes are $\sqrt{15}$ and $-\sqrt{15}$.

Sum of zeroes = $\sqrt{15} - \sqrt{15} = 0 = \frac{-0}{1}$, since the coefficient of t is 0.

Product of zeroes = $\sqrt{15} \times (-\sqrt{15}) = -15 = \frac{-15}{1}$.

Answer: zeroes $\sqrt{15}$ and $-\sqrt{15}$. Verified.

Q1 (vi) Find the zeroes of $3x^2 - x - 4$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$3x^2 - x - 4 = 3x^2 + 3x - 4x - 4 = 3x(x + 1) - 4(x + 1) = (3x - 4)(x + 1)$. The zeroes are $\frac{4}{3}$ and -1 .

Sum of zeroes $= \frac{4}{3} - 1 = \frac{1}{3} = \frac{-(-1)}{3}$.

Product of zeroes $= \frac{4}{3} \times (-1) = -\frac{4}{3} = \frac{-4}{3}$.

Answer: zeroes $\frac{4}{3}$ and -1 . Verified.

Q2 Find a quadratic polynomial whose zeroes have the given sum and product: (i) $\frac{1}{4}, -1$ (ii) $\sqrt{2}, \frac{1}{3}$ (iii) $0, \sqrt{5}$ (iv) $1, 1$ (v) $-\frac{1}{4}, \frac{1}{4}$ (vi) $4, 1$.

SOLUTION

A quadratic polynomial with sum of zeroes S and product of zeroes P is $x^2 - Sx + P$. Any non-zero multiple of it has the same zeroes, so fractions can be cleared.

(i) $x^2 - \frac{1}{4}x - 1$. Multiplying by 4: $4x^2 - x - 4$.

(ii) $x^2 - \sqrt{2}x + \frac{1}{3}$. Multiplying by 3: $3x^2 - 3\sqrt{2}x + 1$.

(iii) $x^2 - 0x + \sqrt{5} = x^2 + \sqrt{5}$.

(iv) $x^2 - x + 1$.

(v) $x^2 + \frac{1}{4}x + \frac{1}{4}$. Multiplying by 4: $4x^2 + x + 1$.

(vi) $x^2 - 4x + 1$.

Answer: (i) $4x^2 - x - 4$ (ii) $3x^2 - 3\sqrt{2}x + 1$ (iii) $x^2 + \sqrt{5}$ (iv) $x^2 - x + 1$ (v) $4x^2 + x + 1$ (vi) $x^2 - 4x + 1$

The questions are restated in MathSetu's own words and the figures are redrawn; refer to the textbook for the original wording. The solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by NCERT, CBSE or any publisher.